

40_Dem_Rep_Congo

October 18, 2025

1 How have female and male contributing family worker rates evolved relative to one another in the Democratic Republic of Congo between 1991 and 2023?

1.1 Abstract

Using World Bank World Development Indicators (WDI), this study examines the evolution of female and male contributing family worker rates in the Democratic Republic of Congo between 1991 and 2023. The analysis compares two key indicators: the percentage of female contributing family workers (% of female employment) and the percentage of male contributing family workers (% of male employment). Over this thirty-two-year period, the data show a marked shift in labor composition: initially, most of the female workforce was engaged as contributing family workers, while male participation was comparatively low. Over time, female rates decreased significantly, while male rates increased moderately. This convergence reflects structural changes in labor dynamics, yet female contribution remains moderately higher than male. These patterns illustrate the gradual balancing of gender participation in family labor and provide insights into employment trends, social transformation, and economic diversification in the DRC.

1.2 1. Question

How have female and male contributing family worker rates evolved relative to one another in the Democratic Republic of Congo between 1991 and 2023?

- **Female proxy:** Contributing family workers, female (% of female employment) (modeled ILO estimate)
- **Male proxy:** Contributing family workers, male (% of male employment) (modeled ILO estimate)

1.3 2. Data

- **Source:** World Bank World Development Indicators (WDI)
- **Indicators:**
 - Contributing family workers, female (% of female employment) (modeled ILO estimate)
 - Contributing family workers, male (% of male employment) (modeled ILO estimate)
- **Coverage:** Democratic Republic of Congo, 1991–2023
- **Notes:** National-level data only

1.4 3. Method

1. Filtered dataset for the Democratic Republic of Congo and selected the two contributing family worker indicators.
2. **Extracted relevant columns:** Year, Indicator Name, and Value.
3. Pivoted the dataset to create a side-by-side comparison of female and male contributing family worker rates between 1991 and 2023.
4. Generated a dual-line time series plot to visualize gender differences, convergence trends, and overall shifts in family labor participation.

(Analysis is descriptive; no causal inference applied.)

1.5 4. Results

- **Female contributing family workers:** Began the period at a high level, reflecting that a majority of women were engaged in family labor. Over time, rates decreased significantly, showing a marked decline in female-dominated family work.
- **Male contributing family workers:** Started at comparatively low levels but increased moderately over the period, indicating gradual male integration into family labor activities.
- **Comparison:** The two indicators show converging trends over time, with female rates still moderately higher than male rates by 2023. The movement reflects a narrowing gender gap in family labor participation, with a clear shift toward more balanced gender distribution.

(Figure 1. Democratic Republic of Congo: Female vs. Male Contributing Family Workers, 1991–2023)

(Table 1. Pivoted dataset summary)

1.6 5. Interpretation

- The declining share of female contributing family workers alongside rising male participation suggests structural shifts in the labor force, possibly linked to urbanization, sectoral diversification, and changing household labor dynamics.
- The convergence of the two lines indicates a gradual reduction of gender disparities in family labor, though women still dominate this category.
- These trends may reflect broader socio-economic transitions, including increased female access to wage employment and male involvement in family enterprises.
- Overall, the data highlight ongoing changes in employment patterns and the evolving role of gender within household and informal labor markets.

1.7 6. Limitations

- National-level aggregates may conceal regional or sector-specific differences, including urban-rural disparities.
- Modeled ILO estimates may contain uncertainty, particularly in earlier years with limited survey coverage.
- The descriptive approach does not determine causal factors behind the decline in female participation or the rise in male contribution, such as policy changes, labor market incentives, or social norms.

1.8 7. Next Steps / Extensions

- Disaggregate by sector (agriculture, informal enterprises) to better understand the contexts driving gendered labor shifts.
- Explore correlations with macroeconomic trends, education levels, and female labor force participation in formal sectors.
- Compare DRC's trends with other Central African countries to examine regional patterns of gendered family labor.
- Conduct time-series analysis to separate long-term trends from short-term fluctuations and identify structural changes in household labor dynamics.

```
[1]: # How have female and male contributing family worker rates evolved relative to one another in the Democratic Republic of Congo between 1991 and 2023?
```

```
import pandas as pd
import matplotlib.pyplot as plt
import os

# Folders
data_raw_folder = "data_raw/"
data_clean_folder = "data_clean/"
figures_folder = "figures/"

# Load CSV
filename = "gender_cod_filtered.csv" # Filtered dataset with only relevant rows
df = pd.read_csv(os.path.join(data_raw_folder, filename))

# Keep only needed columns
df = df[["Year", "Indicator Name", "Value"]]

# Convert Year and Value to numeric, drop invalid rows
df["Year"] = pd.to_numeric(df["Year"], errors="coerce")
df["Value"] = pd.to_numeric(df["Value"], errors="coerce")
df = df.dropna(subset=["Year", "Value"])

# Pivot indicators into separate columns
df_pivot = df.pivot(index="Year", columns="Indicator Name", values="Value").reset_index()
df_pivot = df_pivot.sort_values("Year")

print("Pivoted Democratic Republic of Congo dataset:")
display(df_pivot)

# Interpolate missing values for smooth plotting (optional)
df_plot = df_pivot.interpolate(method='linear')

# Plot the indicators
plt.figure(figsize=(10,6))
```

```

plt.plot(df_plot["Year"], df_plot["Contributing family workers, female (% of
↳female employment) (modeled ILO estimate)"],
        marker='o', linestyle='-', label="Contributing family workers, female
↳(% of female employment)")
plt.plot(df_plot["Year"], df_plot["Contributing family workers, male (% of male
↳employment) (modeled ILO estimate)"],
        marker='o', linestyle='-', label="Contributing family workers, male (%
↳of male employment)")

plt.title("Democratic Republic of Congo: Female vs Male contributing family
↳workers (%) (1991-2023)")
plt.xlabel("Year")
plt.ylabel("Percentage")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(figures_folder,
↳"dem_rep_congo_female_vs_male_contributing_family_workers.png"))
plt.show()

# Save cleaned CSV
df_pivot.to_csv(os.path.join(data_clean_folder,
↳"dem_rep_congo_female_vs_male_contributing_family_workers"), index=False)

```

Pivoted Democratic Republic of Congo dataset:

Indicator Name	Year	\
0	1991	
1	1992	
2	1993	
3	1994	
4	1995	
5	1996	
6	1997	
7	1998	
8	1999	
9	2000	
10	2001	
11	2002	
12	2003	
13	2004	
14	2005	
15	2006	
16	2007	
17	2008	
18	2009	
19	2010	
20	2011	

21	2012
22	2013
23	2014
24	2015
25	2016
26	2017
27	2018
28	2019
29	2020
30	2021
31	2022
32	2023

Indicator Name Contributing family workers, female (% of female employment) 
 (modeled ILO estimate) \

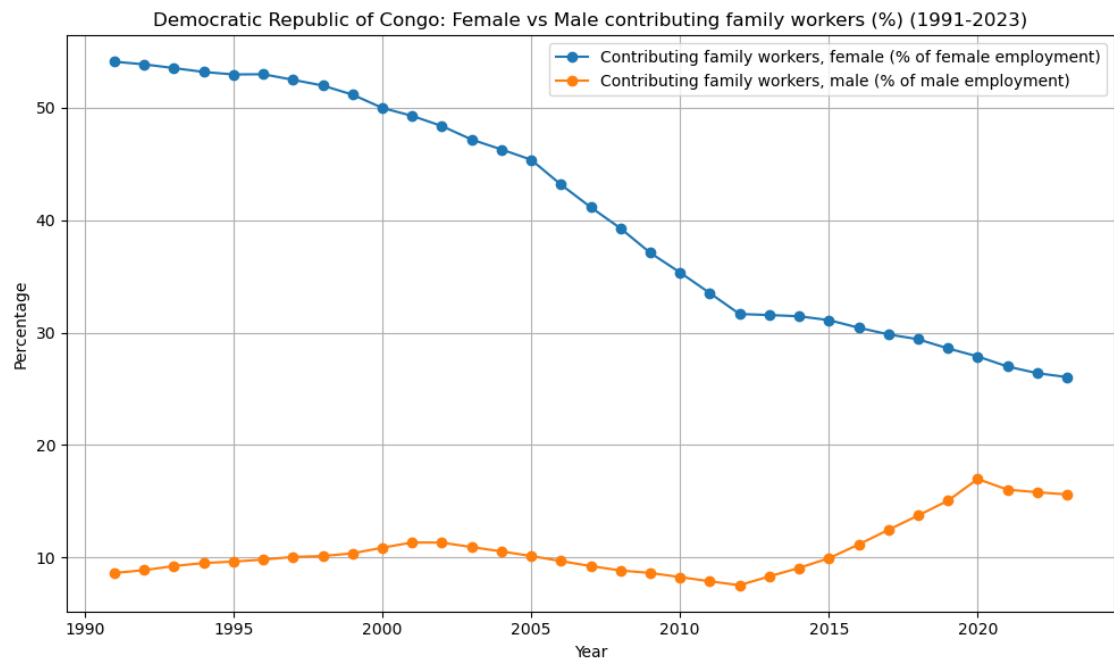
0	54.095501
1	53.856986
2	53.526494
3	53.180763
4	52.959600
5	52.979401
6	52.483327
7	51.972074
8	51.162611
9	49.983505
10	49.276533
11	48.370793
12	47.170881
13	46.290298
14	45.374274
15	43.165957
16	41.151905
17	39.275664
18	37.094936
19	35.348438
20	33.514651
21	31.653853
22	31.561340
23	31.452239
24	31.110393
25	30.434448
26	29.840861
27	29.403367
28	28.593630
29	27.877450
30	26.984339
31	26.388152

32

26.043212

Indicator Name Contributing family workers, male (% of male employment) 
 (modeled ILO estimate)

0	8.620430
1	8.870628
2	9.236439
3	9.507460
4	9.632238
5	9.811763
6	10.047239
7	10.126861
8	10.367797
9	10.866322
10	11.333372
11	11.319680
12	10.924526
13	10.538448
14	10.128448
15	9.677067
16	9.234019
17	8.843839
18	8.620181
19	8.257385
20	7.879642
21	7.530288
22	8.321572
23	9.067221
24	9.932462
25	11.153537
26	12.456281
27	13.733743
28	15.050412
29	16.978183
30	16.030877
31	15.801837
32	15.603504



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